Retail Sales: Where We Make Money and Where We Leak It

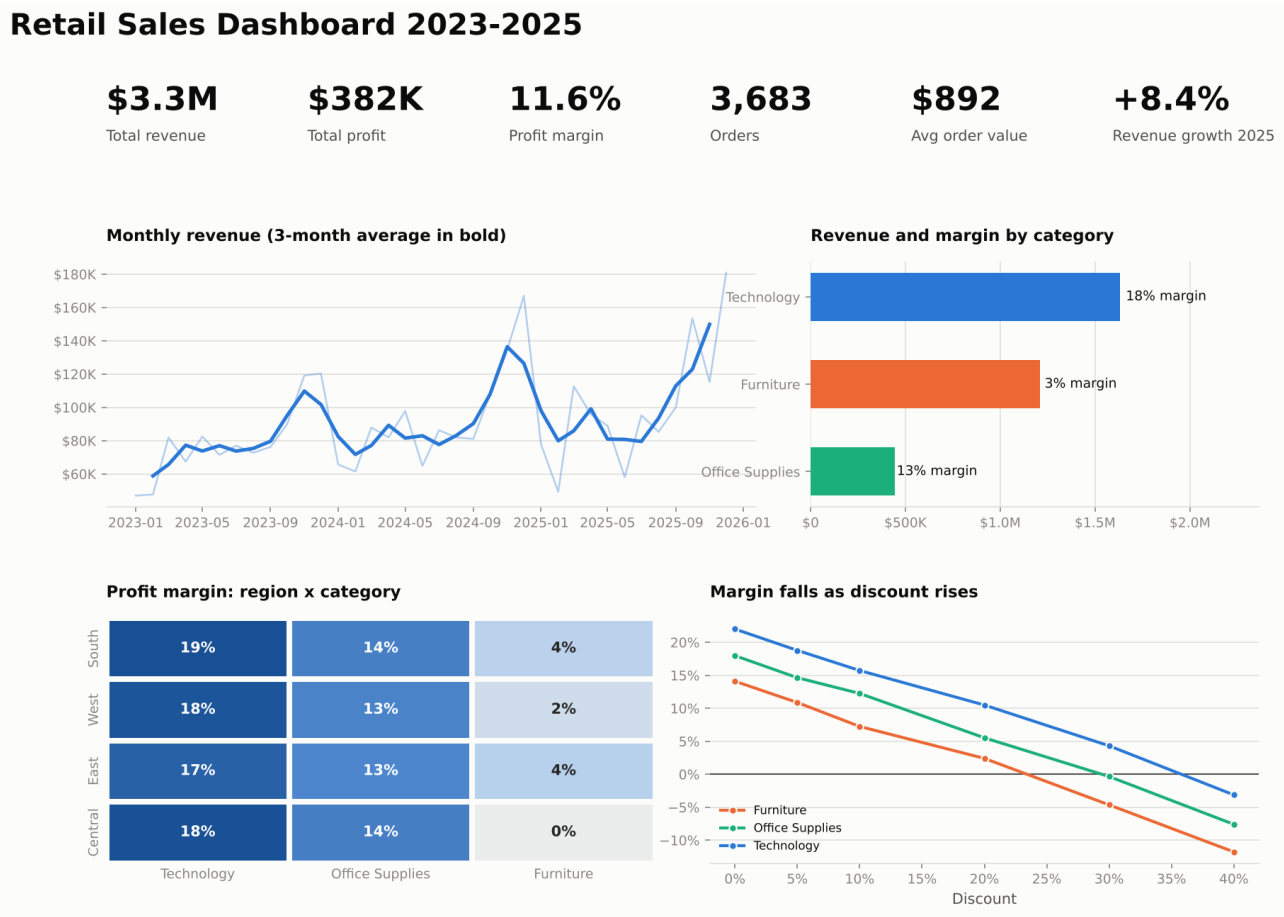
Task 3: Data Visualization | Data story and decision report

$3.3M	$382K	11.6%	3,683	+8.4%
Revenue	Profit	Margin	Orders	Growth 2025

Executive summary

Revenue is growing about 8% a year, with a strong fourth quarter. Profit is not keeping pace. The cause is discounting: orders with a discount of 30% or more earn a **-4.4%** margin, compared with **17.3%** for orders at 10% or less. About 20% of orders carry such discounts, and 18% of all orders lose money. Furniture, and Central Furniture in particular, is where the leak is largest.

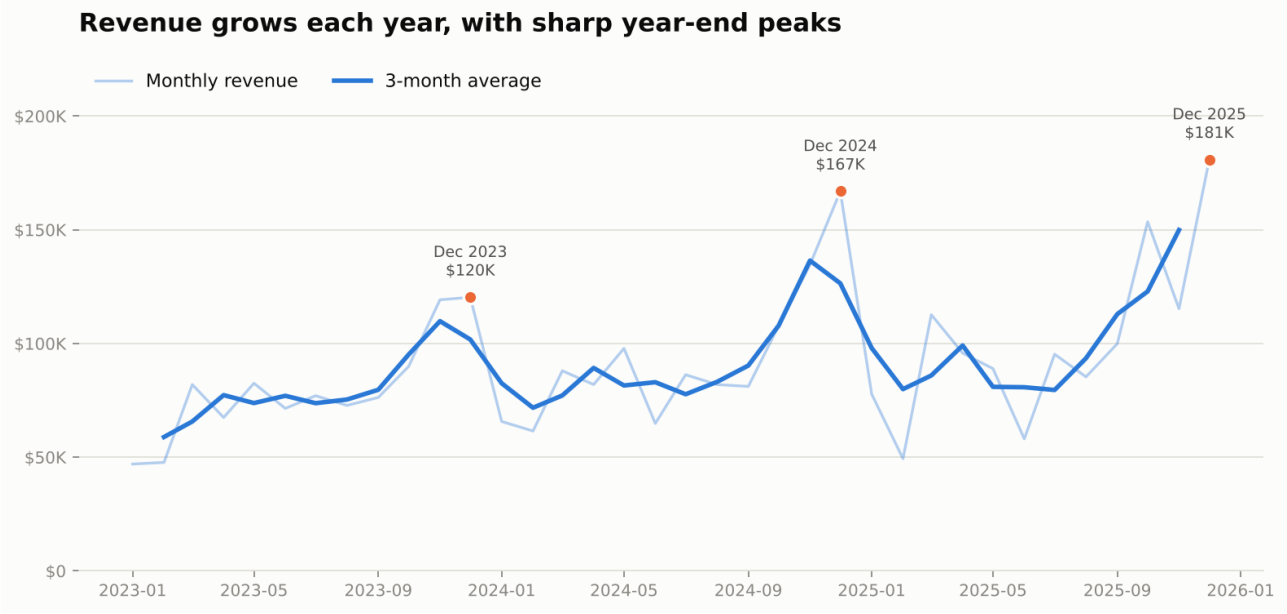
Business question: Where is the business making money, and where is it leaking it?
Data: 3,683 orders, Jan 2023 to Dec 2025 (synthetic dataset with realistic patterns).
Tools: Python, Pandas, Matplotlib, Seaborn.



One-page dashboard: KPIs, revenue trend, category mix, margin heatmap and discount effect.

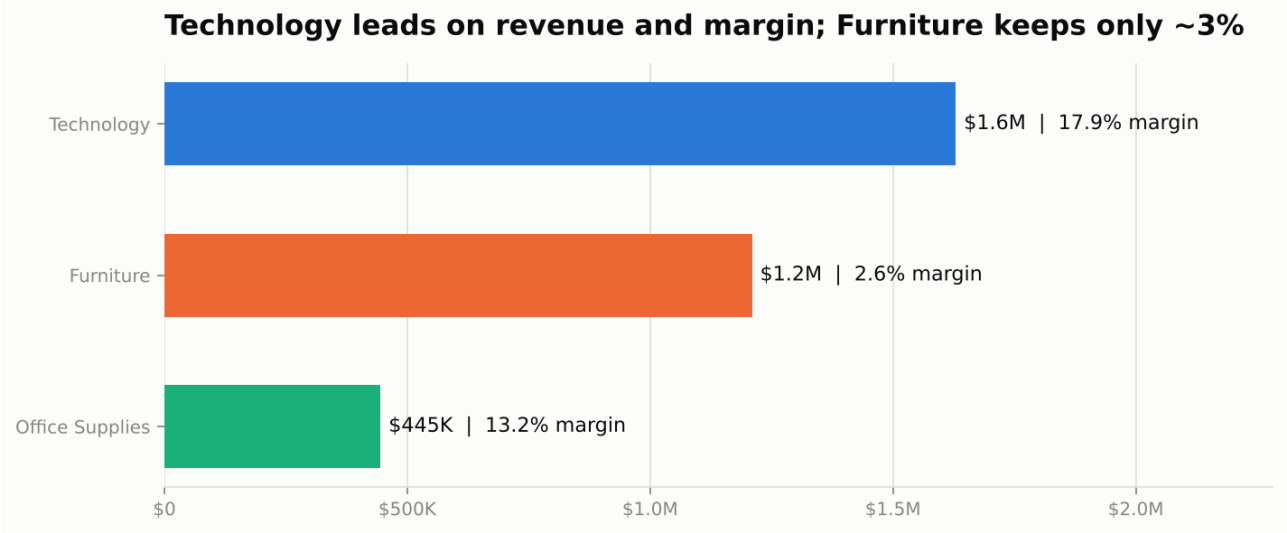
The story in seven charts

1. The business is growing



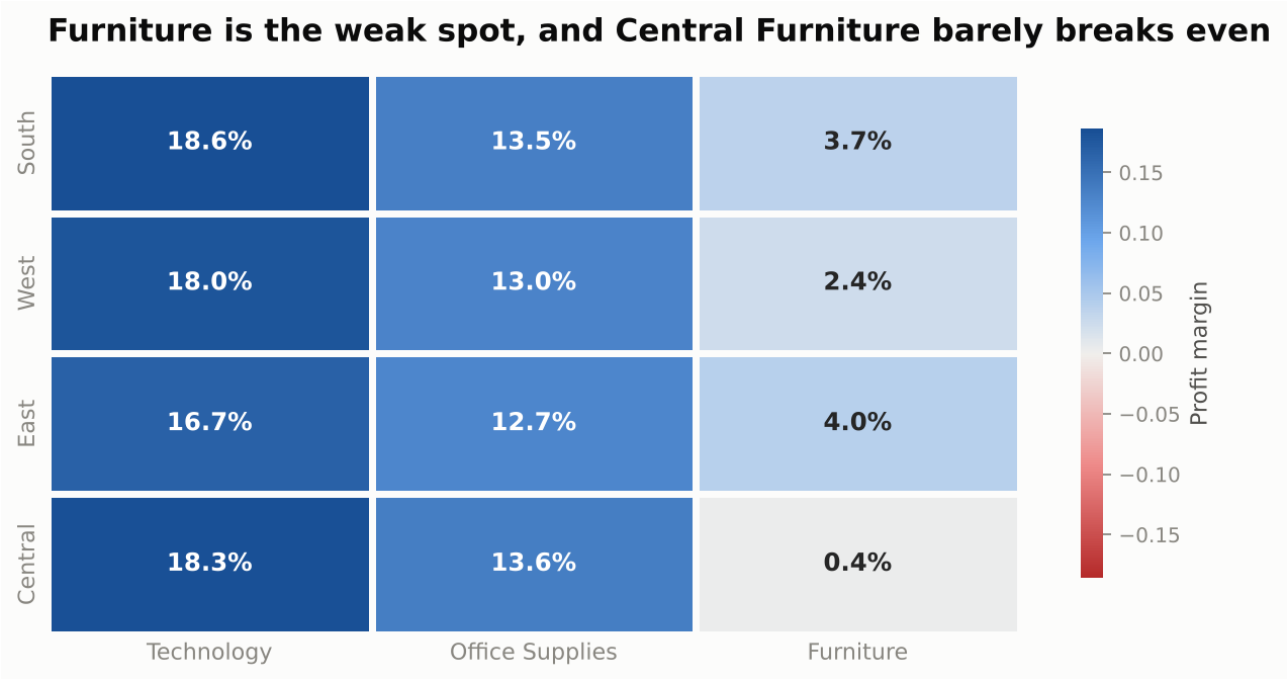
Revenue rose from about $954K (2023) to $1.2M (2025). Every year peaks at year end.

2. Technology wins; Furniture earns little



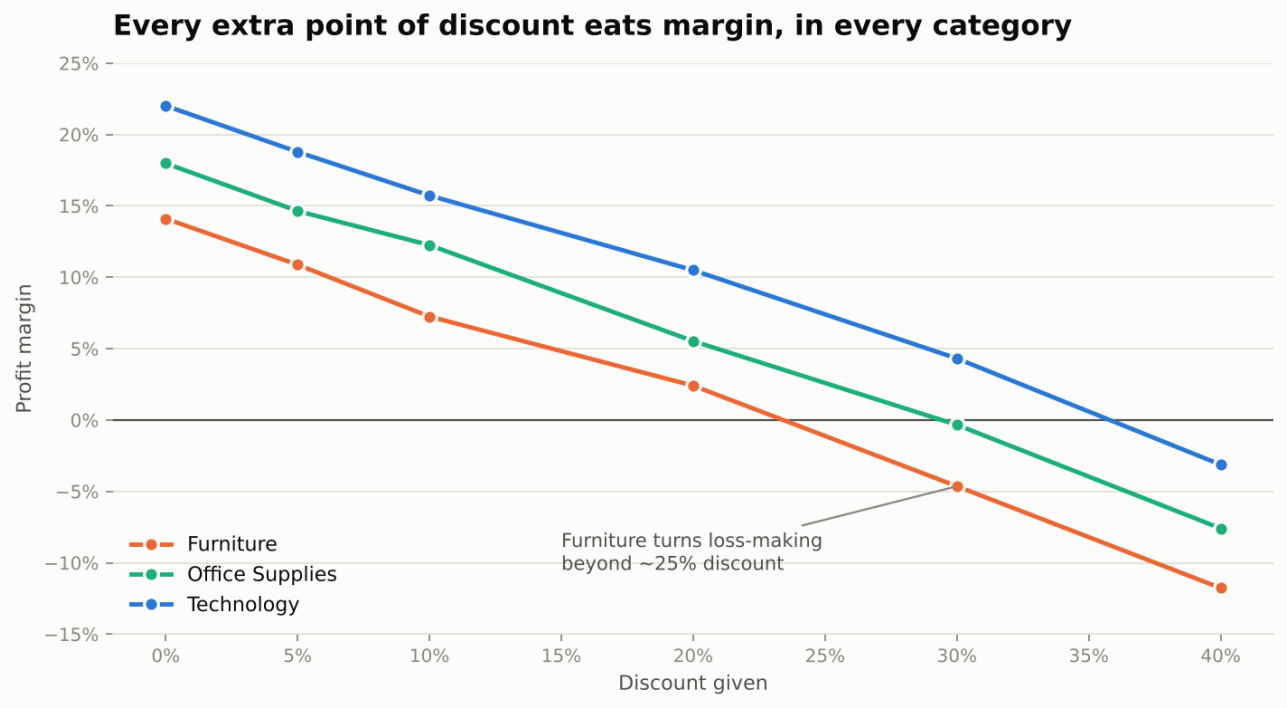
Technology is the largest category and has the best margin (18%). Furniture keeps only about 3% of what it sells.

3. The weak spot is Furniture



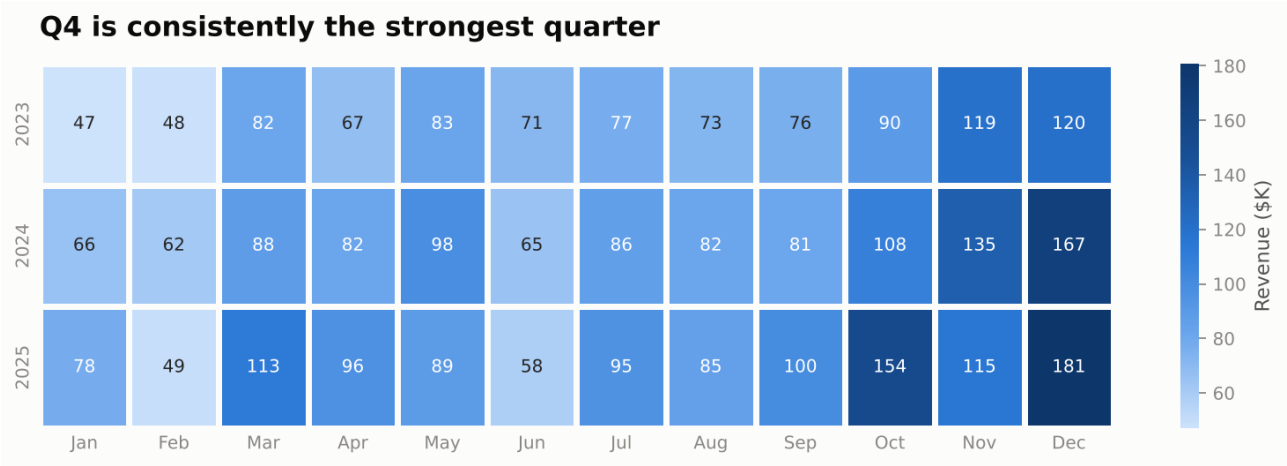
Red would mean a loss. No cell is below zero, but Central Furniture sits at 0.4%, essentially break-even.

4. The cause: discounts



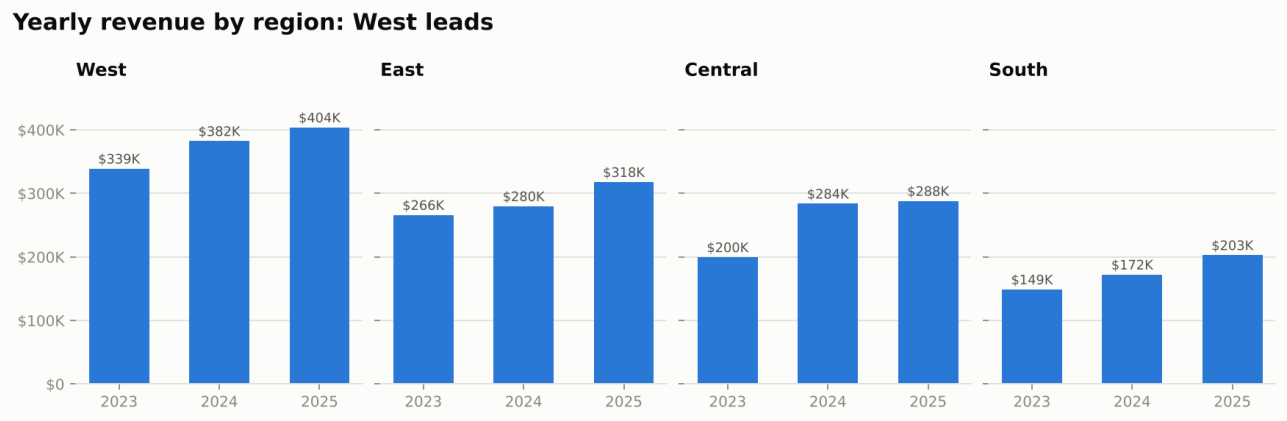
Margin falls steadily as discounts rise, in every category. Furniture turns loss-making beyond roughly 25% discount.

5. Q4 is the money season



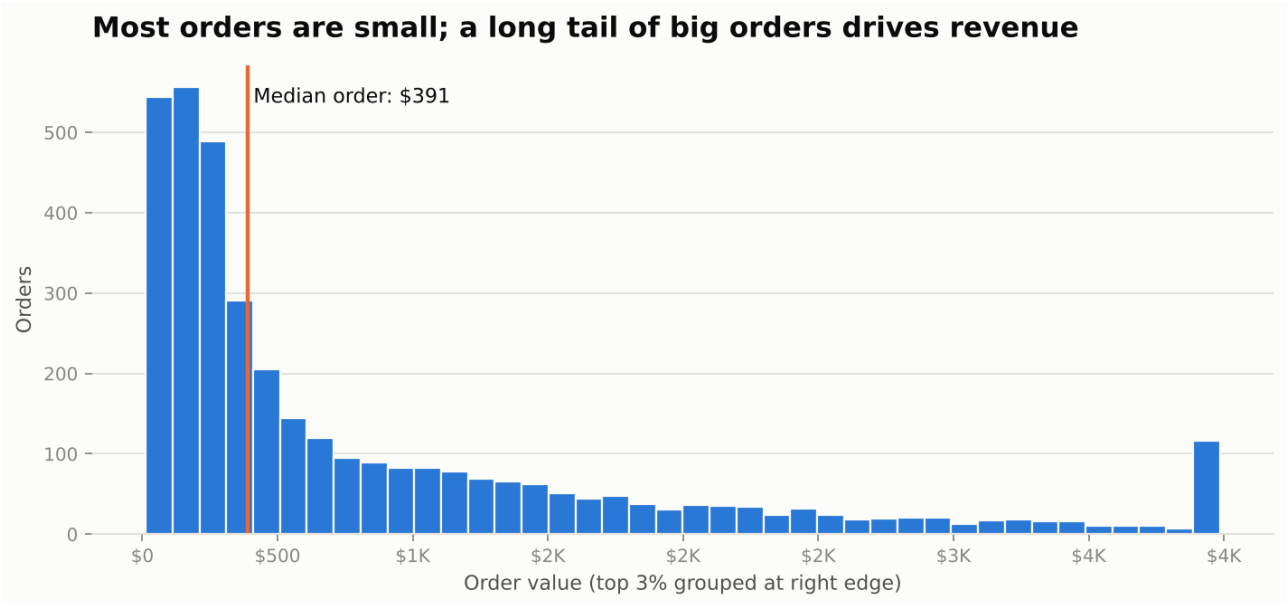
Q4 brings about 36% of annual revenue, every year.

6. Every region grows



West is the largest region. All four regions grew year over year.

7. Most orders are small



Median order is $391, while a long tail of large orders lifts the average to $892.

Recommendations

1. Cap Furniture discounts at about 20% and require approval above that. This is the fastest way to recover margin.

2. Review Central Furniture pricing first. It is the weakest region and category combination.

3. Plan stock and staffing for Q4, when over a third of revenue arrives.

4. Protect Technology margins. It grows well without deep discounts.

Next steps

Pilot the discount cap in one region for a quarter and compare margins against the others. Rebuild the dashboard in Tableau for stakeholders who want to filter by region and category.

Design notes

Each chart title states its finding. Colors are consistent per category (blue, orange, green; colorblind-safe). Sequential scales use one hue for magnitude; the diverging scale uses blue and red around zero for profit. No dual axes; gridlines are light; labels are placed directly on the marks.